

Justin Hart

904-338-8729 | justin.hart@ufl.edu | [linkedin.com/in/justin-b-hart](https://www.linkedin.com/in/justin-b-hart) | justin.hart.science

EDUCATION

University of Florida

Gainesville, FL

Bachelor of Science in Physics and Mathematics, Minor in Computer Science, GPA: 4.0/4.0 August 2024 – May 2028

- **Relevant Coursework:** Quantum Mechanics I, Classical Mechanics II, Thermal Physics I, Mathematical Methods for Physics, Modern Physics, Linear Algebra I, Linear Algebra for Data Science, Real Analysis and Advanced Calculus I, Intro. to Complex Analysis, Intermediate Diff. Eq., Partial Diff. Eq.

St. Johns River State College

Orange Park, FL

Associate in Arts, GPA: 4.0/4.0

August 2022 – May 2024

EXPERIENCE

Undergraduate Research Assistant

December 2025 – Present

University of Florida

Gainesville, FL

- Designed novel pipeline for predicting reduced density matrices using a Neural Network ansatz, enabling accelerated material discovery.
- Implemented method on twisted bilayer MoTe_2 achieving 99.10% accuracy, as compared to 96.40% in standard methods.
- Drafted manuscript titled “Representability-Aware Neural Networks for Reduced Density Matrices: Application to Fractional Chern Insulators” currently in review for *PRX Intelligence*.

NSF Research Intern

May 2026 – July 2026

University of Illinois Urbana-Champaign

Champaign, IL

- Applied novel method for calculating magnon-induced optical spectra towards ferromagnetic FCC Ni.
- Compared method to experimental data, recreating important optical peaks and deducing origin of temperature-induced transitions.

NSF Research Intern

May 2025 – August 2025

Lehigh University

Bethlehem, PA

- Implemented a residual Convolutional Neural Network framework to train 216 autoencoders on 546 Heuslerene band structures, exploring hyperparameters such as latent space dimension, train-test split, and training epochs.
- Applied clustering algorithms (HDBSCAN, K-Means, GMM) to autoencoded fingerprints and designed validation metrics.
- Developed an interactive web application for exploring 2D embeddings.
- Published paper titled “High-Throughput Discovery of Topological and Magnetic Phases in Heuslerenes” in *APL Materials*.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Bash, $\text{\LaTeX}$

Developer Tools: Git/GitHub, Linux, Slurm, VS Code, Anaconda

Libraries: pandas, NumPy, SciPy, Matplotlib, Plotly, Dash, PyTorch, scikit-learn, ROOT

Subjects: Machine Learning, Condensed Matter Physics, Density Functional Theory

AWARDS

UF AI Scholar

March 2026

Florida Academic Top Scholar

March 2025

National Merit Scholar

May 2024

UF Presidential Scholarship Recipient

February 2024

ORGANIZATIONS

High Energy Group Outreach Cabinet Redesign Team

3D Model Lead

Led research on the evolution of particle accelerators and developed 3D CAD models, which were fabricated and painted to create a physical timeline exhibit.

University of Florida Fencing Club

Secretary